

PETERBOROUGH, ON, Canada

WMO: 714360

Lat: 44.2303N

Lon: 78.3526W

Elev: 628

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-4.5	-20.0	1.9	-10.0	-13.6	2.7	-2.9	25.6	26.5	22.1	25.0	3.9	310	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	86.2	72.2	83.3	70.4	80.7	68.8	74.6	82.7	72.7	80.1	71.0	77.7	9.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	121.4	79.0	70.3	114.1	76.7	68.7	107.9	75.0	38.5	82.9	36.8	80.1	35.3	77.8	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.1	15.9	DB	-19.1	90.7	7.0	4.0	-24.1	93.6	-28.2	96.0	-32.1	98.2	-37.2	101.1	(4)
(5)			WB	-19.5	78.1	6.9	2.0	-24.5	79.5	-28.5	80.7	-32.4	81.8	-37.4	83.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	18.5	20.5	29.5	42.6	54.9	63.6	67.8	66.1	58.8	47.2	35.8	25.6	(6)	
	DBStd	19.38	12.63	11.18	10.84	8.09	7.60	6.30	5.26	5.29	7.22	7.86	8.53	10.44	(7)	
	HDD50	4069	977	825	639	246	36	1	0	0	10	151	428	756	(8)	
	HDD65	7824	1442	1245	1101	671	326	100	30	51	210	552	875	1221	(9)	
	CDD50	2015	0	0	3	25	187	408	552	498	275	64	3	0	(10)	
	CDD65	295	0	0	0	0	12	57	117	84	24	1	0	0	(11)	
	CDH74	3092	0	0	1	11	183	604	1190	801	287	15	0	0	(12)	
	CDH80	709	0	0	0	0	32	140	306	173	58	0	0	0	(13)	
(14) Wind	WSAvg	6.6	7.5	7.5	7.7	7.9	6.6	5.8	5.5	4.9	5.0	6.4	6.7	7.3	(14)	
(15) Precipitation	PrecAvg	35.0	2.8	2.0	2.2	3.0	3.3	3.2	3.3	2.7	3.1	2.9	3.2	3.0	(15)	
	PrecMax	46.6	5.2	3.9	4.0	5.7	5.2	7.7	9.8	5.4	6.8	6.6	6.0	5.2	(16)	
	PrecMin	27.5	1.6	0.8	0.7	1.1	0.9	1.1	1.3	1.1	1.5	1.0	1.9	0.7	(17)	
	PrecStd	4.8	0.9	0.9	1.0	1.2	1.1	1.8	1.8	1.1	1.2	1.4	1.1	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	49.5	67.7	75.1	84.0	87.3	90.0	89.5	86.2	76.2	62.8	53.7	(19)	
		MCWB	48.0	45.1	57.2	61.5	69.6	71.2	75.2	72.7	72.8	66.6	54.4	51.2	(20)	
	2%	DB	43.1	43.4	56.0	68.3	79.0	84.0	86.6	84.4	81.0	70.2	57.2	47.1	(21)	
		MCWB	40.8	40.0	48.9	55.0	65.9	70.7	73.0	71.1	69.6	61.4	53.0	44.8	(22)	
	5%	DB	38.8	39.1	49.6	62.6	74.7	80.8	83.3	81.2	77.1	65.6	53.6	42.9	(23)	
		MCWB	36.9	36.3	43.5	51.5	62.2	68.9	70.5	69.2	67.1	59.1	49.1	40.7	(24)	
	10%	DB	35.4	36.0	44.6	57.6	70.4	77.3	80.5	78.6	73.1	61.1	49.7	38.9	(25)	
		MCWB	33.6	33.7	39.6	48.2	59.6	66.3	68.9	67.9	64.9	56.0	46.0	36.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	45.4	58.5	62.7	71.3	75.3	77.9	76.3	74.3	68.3	56.7	51.5	(27)	
		MCDB	49.7	47.9	65.3	71.9	81.1	84.3	86.9	83.6	82.2	73.1	59.4	52.9	(28)	
	2%	WB	41.3	40.6	49.7	57.8	67.9	72.6	75.1	73.6	71.8	63.2	53.5	45.4	(29)	
		MCDB	42.6	42.7	54.1	65.4	76.9	81.5	83.3	80.6	78.3	68.3	56.6	46.8	(30)	
	5%	WB	37.2	36.6	44.3	53.3	64.6	70.5	72.8	71.9	69.1	59.8	49.7	41.0	(31)	
		MCDB	38.9	38.6	49.1	60.4	71.9	77.8	80.0	78.3	74.6	64.4	52.9	42.8	(32)	
	10%	WB	33.7	33.9	40.0	49.4	61.3	68.3	70.8	70.0	66.5	56.7	46.1	37.0	(33)	
		MCDB	35.1	35.7	44.1	56.2	68.0	74.3	77.5	75.7	71.6	60.3	49.4	38.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	18.1	18.0	19.8	22.2	21.4	22.4	22.4	23.3	19.1	15.9	14.1	(35)	
		MCDBR	15.4	19.0	25.1	30.1	28.5	26.1	25.5	26.1	27.8	26.2	22.9	16.5	(36)	
	5% WB	MCWBR	14.5	16.4	18.0	18.7	15.7	14.3	13.4	14.2	16.8	18.1	18.3	15.0	(37)	
		MCWBR	15.7	17.6	23.4	25.4	24.9	22.4	22.1	21.6	23.7	23.7	20.9	15.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.303	0.305	0.319	0.355	0.378	0.413	0.413	0.403	0.365	0.351	0.309	0.283	(40)	
		taud	2.326	2.329	2.419	2.381	2.361	2.284	2.295	2.315	2.414	2.442	2.509	2.455	(41)	
	Ebn at Noon	Ebn at Noon	257	280	292	288	283	272	271	270	272	261	257	257	(42)	
		Edh at Noon	25	30	32	36	38	41	40	38	32	27	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	424	704	1119	1412	1714	1840	1876	1640	1309	768	475	340	(44)	
	RadStd	RadStd	44	64	104	151	181	149	139	83	114	71	60	42	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page